From Codes to Causes: A Causal Interpretability Audit of Quantized Delta Representations in a Transformer World Model

R. Xia[1*]

[1*]Lancaster University, Lancaster, LA1 4YW, United Kingdom.

Corresponding author(s). E-mail(s): r.xia2@lancaster.ac.uk;

Abstract

World-model agents compress environment dynamics into discrete latent codes, but whether those codes form an interpretable, causally meaningful vocabulary has not been tested. We audit a single fully trained **Δ**-IRIS agent on Crafter across descriptive, informational and interventional evidence. The tokenizer's context-aware "delta" codes form a near-monosemantic event vocabulary—single codes predict specific achievement unlocks with conditional probability up to **0.92** and lift up to **708**×—and a controlled ablation isolating the delta conditioning shows this is what sharpens them. Linear probes expose a confound: cumulative agent state is read off the on-screen heads-up display, not transformer memory. Inside the imagination loop, projecting out one sparse-autoencoder direction suppresses an event's codes, whereas perfectly decoding probe directions leave rollouts unchanged; stacking sixteen probe directions never matches the dictionary basis, though the concept stays fully decodable after their removal. The audit offers a transferable methodology and cautions against equating decodability with mechanism.

Keywords: world models, interpretability, reinforcement learning, discrete representations, sparse autoencoders, causal analysis

Model-based reinforcement-learning agents increasingly learn inside their own imagination: a learned *world model* predicts future observations, rewards and terminations, and the policy is optimised largely on imagined trajectories [1–4]. A prominent family tokenizes observations into discrete latent codes and models dynamics autoregressively with a transformer [5]. The recent Δ-IRIS agent [6] refines this with *context-aware* *tokenization*: rather than encoding each frame in isolation, its tokenizer encodes the *transition* (x_t, a_t, x_{t+1}) into four discrete codes, and its decoder reconstructs x_{t+1} with direct access to (x_t, a_t). The codes are thereby pressured to carry only what is *new*—a compressed description of the frame-to-frame delta.

This suggests an interpretability hypothesis: if codes describe changes, and meaningful events are changes, the codebook should be an *event vocabulary*—discrete, enumerable, potentially human-legible. Whether this holds, and more importantly whether such codes (or representations derived from them) are *causally* implicated in the model's predictions rather than merely correlated with events, has to our knowledge not been tested. The distinction matters because high decoding accuracy does not imply a representation is used [7–9]: interpretability claims about world models should climb from description, through information, to intervention before they are believed.

We test the hypothesis on one fully trained Δ-IRIS agent (one seed, one environment) on Crafter [10], a procedurally generated survival game with 22 ground-truth achievement signals that fire at known timesteps—ideal event labels. We audit it in three tiers of evidentiary strength: *descriptive* (codebook statistics and code–event association over the full 10^7-step replay buffer); *informational* (linear probes and concept activation vectors [11] across the tokenizer, the convolutional frame encoder and every transformer block, plus a TopK sparse autoencoder [12, 13] on the residual stream); and *interventional* (projection ablation on single-step prediction and inside the multi-step imagination loop, multi-direction subspace ablation, and activation patching [14, 15]). This audit is, to our knowledge, the first systematic interpretability study of the discrete *transition* vocabularies learned by IRIS-family world models; it complements interpretability-by-architecture approaches [16] and post-hoc analyses of game-playing and next-token sequence models, where probe-decoded world-state directions are causally steerable [17–20].

Four findings result. (i) The delta codes form a near-monosemantic event vocabulary, and a controlled baseline shows the *delta conditioning* causes it. (ii) A "heads-up-display (HUD) shortcut": probes suggest the transformer tracks episodic state, until a control representation—the frame encoder alone—matches it, because Crafter renders state on screen. (iii) A *read–write asymmetry*: supervised probe directions decode at near-perfect AUROC yet are causally inert, while an unsupervised dictionary direction is causally load-bearing inside imagination; the asymmetry persists under rank-16 subspace ablation. (iv) Next-code prediction is causally quasi-Markovian. Beyond the specific results, the audit (Fig. 1)—descriptive association, probing with shortcut controls, dictionary learning, then intervention inside imagination—is a transferable methodology for interrogating discrete world-model representations.

1 Results

1.1 Delta codes form an event vocabulary

Re-encoding the agent's entire replay buffer (29,136 episodes; 9,980,073 transitions) shows no codebook collapse—all 1024 codes are used—but heavily concentrated

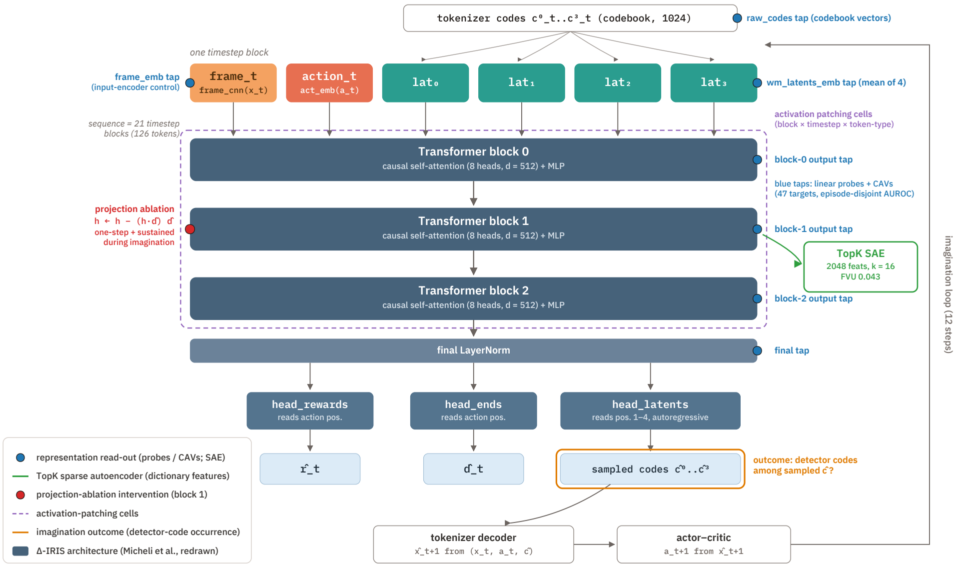

Fig. 1 Overview of the audit. Grey-blue: the Δ-IRIS world model of Micheli et al. [6] (redrawn). Coloured: our instrumentation—blue, the seven probe/CAV read-out points including the frame_emb control; green, the TopK sparse autoencoder on the block-1 output; red, the projection-ablation point; purple, the activation-patching cell grid; orange, the imagination-loop outcome.

usage (per-slot entropy 5.3–6.0 of 10 bits; the single code 29 dominates every slot and marks "no change in this quadrant") (Supplementary Fig. S1). The informative vocabulary lives in the long tail. Computing, for every (slot, code, achievement) cell, the conditional probability $P(u^i \mid c^k{=}j)$ and its lift over base rate, the strongest codes are bona-fide event detectors: (s_3, c_{75}) co-occurs with `collect_iron` 88% of the time (708× chance), (s_3, c_{938}) with `place_plant` at 0.92 (543×), (s_2, c_{707}) with `collect_sapling` at 0.90 (Supplementary Table S1; all maxima $p < 10^{-40}$). Slot 3 hosts 11 of 18 per-achievement maxima, indicating specialisation beyond spatial decomposition. Some codes are polysemantic superclasses: (s_3, c_{36}) detects `place_table`, `make_wood_pickaxe` and `make_wood_sword`—events whose visual signature (a wooden item appears) is near-identical—a collapse the dictionary later resolves. Combat events are the weakest (`defeat_skeleton`: best $P{=}0.07$), plausibly because they look heterogeneous.

The delta conditioning causes the sharpening. Is this monosemanticity due to context-aware delta tokenization specifically, or to discrete quantization in general? Holding the scheme fixed and varying only the conditioning, we train two baselines on the same buffer and run the identical analysis (Methods): a *frame-only ablation* of the Δ-IRIS tokenizer (same backbone, codebook and token count, but encoding x_{t+1} alone) and the *original IRIS* frame tokenizer [5]. The delta codes are ≈ 7× stronger per-event detectors (mean best $P(a \mid c)$ 0.39 vs 0.06 and 0.06; five achievements with

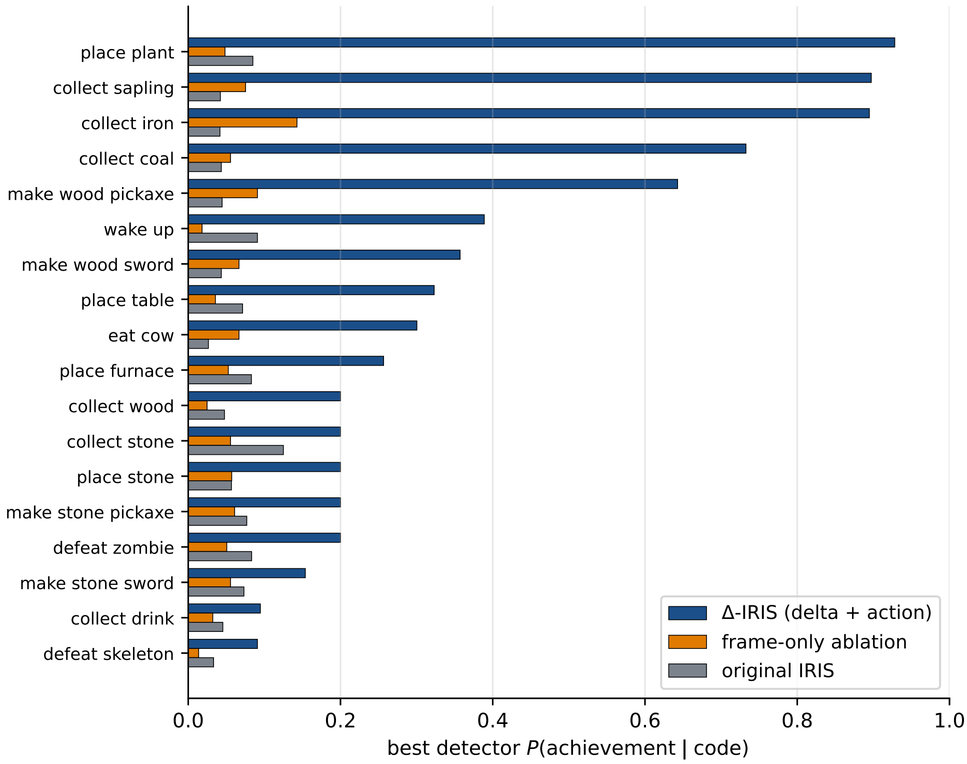

Fig. 2 Context-aware delta tokenization is what turns the codebook into an event vocabulary. Per-achievement best detector $P(\text{achievement} \mid \text{code})$ for the Δ-IRIS delta tokenizer, its frame-only ablation, and original IRIS, under identical analysis on a common 150-episode labelled set. Δ-IRIS dominates on every achievement; the frame-only ablation tracks literal IRIS, isolating the gain to the delta+action conditioning rather than codebook size, token count or architecture.

a strong $P \geq 0.5$ detector vs zero), and Δ-IRIS dominates on every achievement individually (Fig. 2). Critically the two frame baselines land almost on top of each other despite differing codebook sizes, token counts and architectures: the gap is attributable to the delta+action conditioning, not to incidental differences. Both baselines are healthy tokenizers (PSNR 18.4/22.5, broad codebook use; Methods), so the gap is not a degenerate-baseline artefact.

1.2 A HUD shortcut masquerades as memory

We train logistic-regression probes for 47 targets on seven representations spanning the tokenizer codes, the world model's convolutional frame embedding (frame_emb; no transformer), its code embedding, and the four transformer stages, with episode-disjoint splits (Fig. 3; Methods). Just-unlocked events are decodable from the raw codes alone (AUROC ≥ 0.85 for all 18 measurable achievements, ≥ 0.95 for 14), confirming the association analysis per-sample. Cumulative state ("has the agent ever collected wood?") tells a subtler story: near chance from the codes, but excellent (0.84–1.00) from any transformer block—inviting the conclusion that the transformer maintains episodic memory. The input-encoder control refutes it: frame_emb, seeing only the current frame, *matches or exceeds every transformer block on 16 of 18* cumulative concepts, a margin that survives paired-bootstrap 95% confidence intervals (Methods). Crafter renders the inventory in the HUD, so cumulative state is written on the screen and one attention step relays it. The one probed target where depth helps monotonically is short-horizon reward anticipation (reward_in_next_5: $0.77 \to 0.81$ across blocks), the cleanest evidence of what the world model's depth adds. We argue this input-encoder control should be standard when probing pixel-based RL agents, whose observations routinely render persistent state [8, 21].

The probe weight vectors are concept activation vectors (CAVs); their pairwise geometry recapitulates Crafter's tech tree (most-aligned pairs are tech-tree neighbours, e.g. cos(cum[collect_stone], cum[place_stone])=0.95), while event and state directions for the same achievement are weakly anti-aligned—though within-episode co-occurrence inflates these alignments, and the intervention tier shows these directions are read-out only. A TopK sparse autoencoder on the block-1 residual stream (Methods) reconstructs well (fraction of variance unexplained 0.043; zero dead features) and is stable across widths, sparsities, seeds and layers (Methods). It splits the polysemantic code (s_3, c_{36}) into *distinct near-monosemantic features* for place_table, make_wood_pickaxe and make_wood_sword: the continuous residual stream retains distinctions the tokenizer's 40-bit bottleneck collapses, and dictionary learning recovers them (Supplementary Table S2).

1.3 Dictionary directions write; probe directions read

Decodability does not imply mechanism. We measure the causal contribution of a direction by projection ablation, $h' = h - (h^\top \hat{d})\hat{d}$, on the block-1 output, contrasting unlock against matched ordinary moments and matched against random directions (Methods). Ablating the matched SAE direction at the unlock step costs up to +4.85 nats (wake_up), with collect_coal (+2.41) and collect_sapling (+1.73) close behind; eight of 18 concepts are significant but the effect concentrates in a handful. Random directions are inert, and an empirical null of 50 random directions confirms specificity (matched effect exceeds the 99th percentile for collect_coal z=24, collect_sapling z=9.9, make_stone_pickaxe z=3.1; honest negatives such as collect_iron fall below the null; Supplementary Table S6). Ablating three steps earlier eliminates the effect entirely—the un-ablated frame embeddings re-inject the information within three steps, the interventional face of the HUD shortcut (Supplementary Fig. S3).

The stronger test runs inside imagination: after an 8-step burn-in we let the agent dream 12 steps with a direction continuously projected out, and ask whether the event's detector codes are sampled (Fig. 4). For collect_wood, the baseline dreams the wood code at the first imagined step in 100% of rollouts; ablating SAE feature f_{187} reduces this to 0% (Fisher $p < 10^{-9}$), while the CAV and a random direction leave it at 100%. Removing a single direction in the 512-dimensional residual stream suppresses an event class in imagined futures; decoded filmstrips confirm the event vanishes

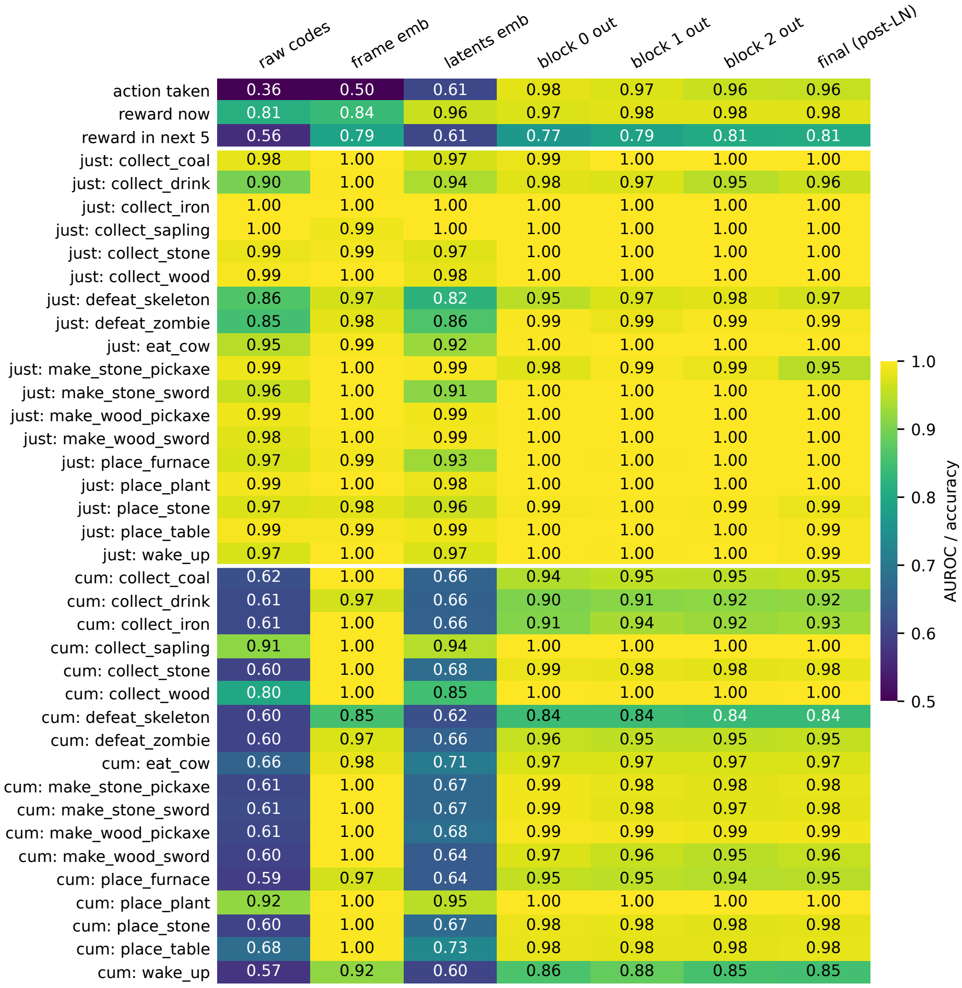

Fig. 3 The HUD shortcut. Held-out probe AUROC (accuracy for the 17-way action probe) for 39 measurable targets across seven representations, episode-disjoint, n=81,919 steps. frame_emb is the input-encoder control (a CNN over the current frame, no transformer). Cumulative-state rows are answered by frame_emb alone; only reward_in_next_5 improves materially with transformer depth.

under the matched ablation but survives an off-target or random one (Supplementary Fig. S4). Across *all* concepts, ablating the supervised CAV direction—near-perfect AUROC—changes the outcome by essentially nothing (max $|\Delta|$=0.10): the directions that suffice to *read* a concept are not those the computation *writes through*. Adding the SAE direction (steering) only weakly induces events (mean +0.04), so the dictionary direction is closer to necessary than sufficient, and this agent shows no activate-over-suppress asymmetry (Methods).

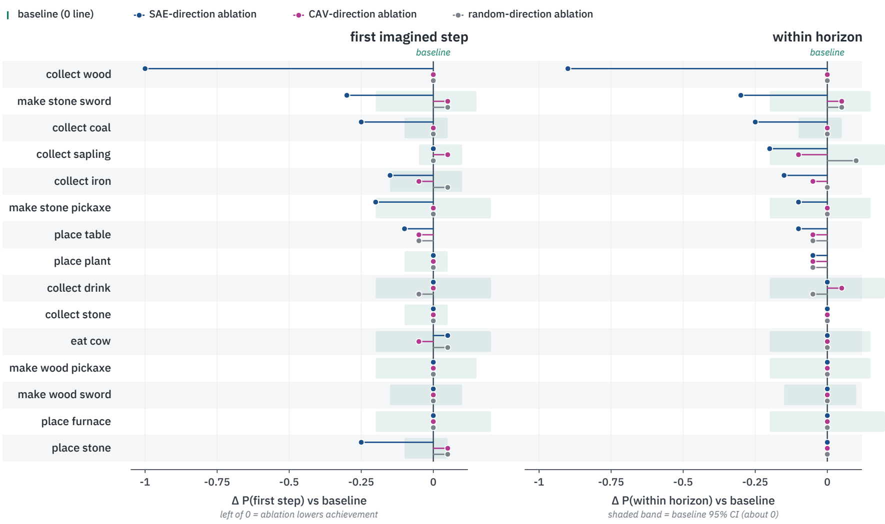

Fig. 4 Read–write asymmetry inside imagination. Change in the probability that a concept's detector code is sampled during 12-step imagination—first step (left) and anywhere in horizon (right)—under SAE-direction (blue), CAV-direction (magenta) and random-direction (grey) ablation, relative to baseline; n=20 rollouts. Shaded band is the baseline 95% CI. The matched SAE direction selectively lowers achievement (dramatically for `collect_wood`); CAV and random stay within noise.

The asymmetry survives multi-direction ablation. Single-direction removal is a weak instrument against redundant codes [22]. Generalising to a rank-r orthonormal subspace $h' = h - QQ^\top h$ (Methods), the matched SAE-feature stack accumulates causal weight with rank—mean $\Delta \log P$ climbs 0.51 $\to$ 2.57 nats for r=1 $\to$ 16 (up to +6.3 for `collect_coal`)—whereas a matched CAV subspace saturates near 0.5 and even declines (0.48 $\to$ 0.32), an $\approx 8\times$ gap at r=16 (Fig. 5A). Yet a probe refit after projecting out the rank-16 CAV subspace still decodes at AUROC $\approx$ 0.999 (Fig. 5B): the decodable information is massively redundant, distributed far outside any low-rank CAV span. Direct logit attribution corroborates the writing role—the SAE direction's mean absolute contribution to its detector code's logit is 0.31, against 0.08 (CAV) and 0.01 (random), and is detector-specific (Methods). The headline thus upgrades from "one SAE direction sufficed where one probe direction did not" to: *the dictionary basis, not the probe basis, is what the computation writes through.*

1.4 Next-code prediction is quasi-Markovian

Localising the causal signal by activation patching—restoring clean activations one (block, timestep, token-type) cell at a time into a corrupted run (Methods)—the picture is strikingly concentrated (Fig. 6). Patching historical cells restores almost nothing (mean $|\leq 0.04|$ over the earliest 15 steps); at the final step the action token is the largest carrier (restoration 0.46–0.58), with the latent tokens dominant for two of four concepts. Next-code prediction is computed from the current step's action and frame, not from transformer memory of the preceding twenty steps—a third line of

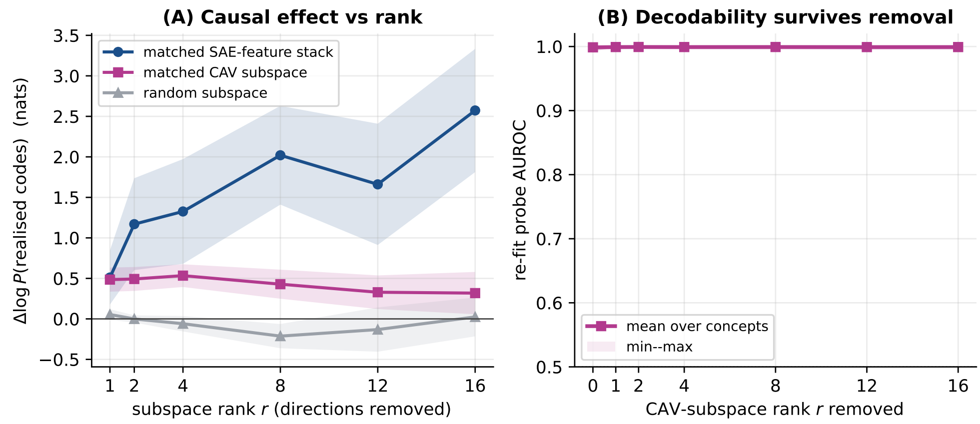

Fig. 5 Multi-direction ablation. (A) mean $\Delta \log P$ of the realised codes (over eight dominant concepts; band $\pm$SEM) when a rank-r subspace is projected out at block 1: the matched SAE-feature stack climbs with rank, the matched CAV subspace saturates ($\approx 8\times$ smaller at r=16), random is inert. **(B)** a probe re-fit after removing the rank-r CAV subspace still decodes at AUROC ≈ 0.999—decodability is redundant and is not what the model writes through.

evidence, after the frame_emb control and the gap-3 ablation recovery, for the same conclusion.

2 Discussion

The three tiers compose into a consistent functional anatomy of Δ-IRIS on Crafter. The tokenizer's delta codes are the event vocabulary—discrete, sparse and, in the tail, near-monosemantic—and a controlled ablation shows the context-aware delta conditioning, not quantization per se, is what sharpens them. The convolutional frame encoder carries persistent state, not because the architecture earmarks it for memory but because the environment renders state on screen; the transformer's distinctive contribution is anticipation, while its apparent episodic memory dissolves under controls. Within the residual stream, the directions that causally carry events are the sparse dictionary directions, not the directions that best decode them.

Two methodological lessons are cheap to adopt and did the heaviest lifting. First, the input-encoder probe as a shortcut control: without it we would have reported, wrongly but at AUROC 0.98, that the transformer tracks episodic state. Second, the double contrast in ablation (matched vs ordinary moments × matched vs random directions), without which single-direction ablations are uninterpretable. More broadly, descriptive and informational evidence—association tables, probe accuracies, concept geometries—should be treated as hypothesis generators for intervention, not as findings about mechanism [7, 21]. The read–write asymmetry has a plausible mechanism: a probe is free to exploit any correlated signal across many directions, whereas a TopK autoencoder must reconstruct the activation from few directions, so its decoder

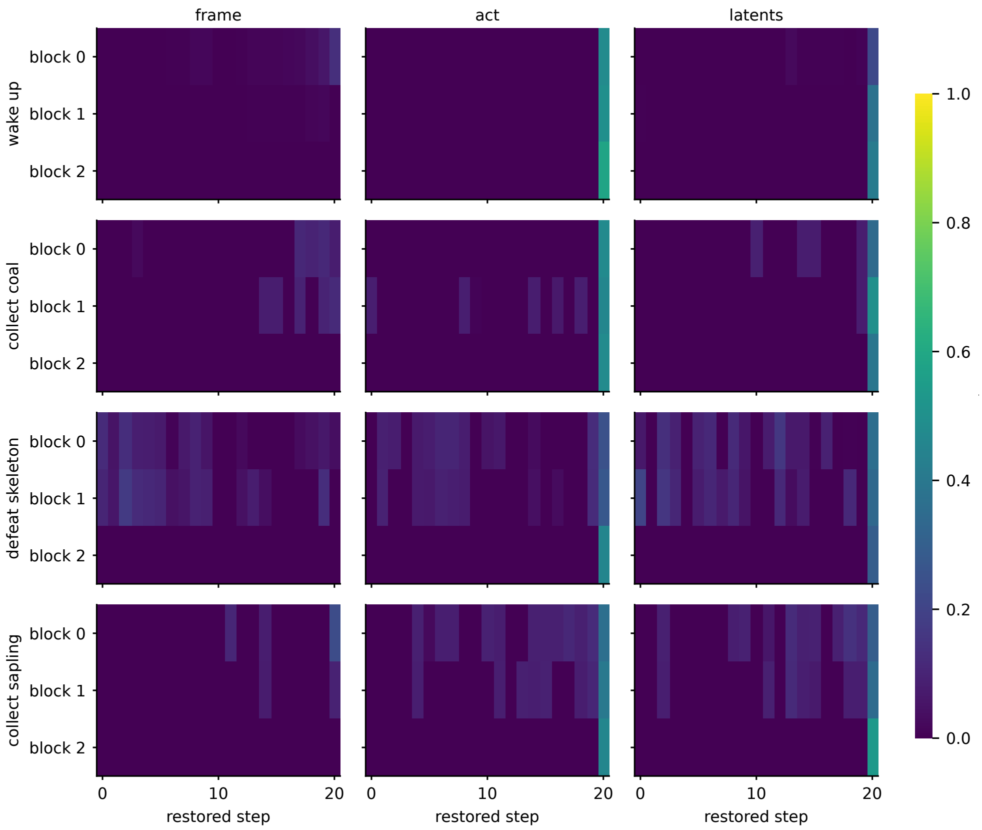

Fig. 6 Causal tracing. Mean restoration of unlock-code prediction when clean activations are patched into a corrupted run, per (block × timestep) cell, for frame, action and latent token positions (columns) and four concepts (rows). Restoration concentrates at the final timestep on the action and latent tokens; historical cells restore essentially nothing.

columns span the signal the computation consumes [12, 23]; recent reports that linear concept-removal and steering carry unintended effects [24–26] are consistent with this. The quasi-Markovian finding is double-edged: it shows economy (the model does not duplicate on-screen state in attention) but implies Crafter underuses the 21-step context, so environments with genuinely hidden state would better exercise—and let one audit—the transformer's memory.

Our audit covers one agent, one seed, one environment, and its central HUD finding also limits how much the transformer must remember; the audit should be repeated in partially observed environments without on-screen state. We extend the read–write asymmetry from single directions to rank-16 subspaces, but larger subspaces, other layers and iterative nullspace projection remain; our sufficiency (steering) test was only weakly positive; and the imagination outcome measure shares provenance with the feature-selection statistic, a circularity we flag throughout. With those caveats, the discipline the audit recommends generalises: in world models as in language models, interpretability claims should climb from description to intervention before they are believed.

3 Methods

3.1 The Δ-IRIS agent

Δ-IRIS [6] comprises a tokenizer, a transformer world model and a CNN–LSTM actor–critic trained jointly on imagined trajectories. The tokenizer encodes a transition (x_t, a_t, x_{t+1}), $x \in \mathbb{R}^{3\times64\times64}$, by stacking both frames and an action-embedding channel through a convolutional encoder; the latent is reshaped into a 2×2 grid of slots, each vector-quantised [27] against a shared codebook of C=1024 entries (d=64, cosine), giving four codes $\mathbf{c}_t \in \{0, \ldots, 1023\}^4$. The decoder reconstructs x_{t+1} from $(x_t, a_t, \mathbf{c}_t)$, so the 40 bits of code need carry only the transition's residual novelty. Each timestep contributes a six-token block (frame embedding, action embedding, four code embeddings) to a causal transformer (L=3 blocks, 8 heads, d_{model}=512, context 21 timesteps). Three MLP heads read fixed intra-block positions: `head_rewards`/`head_ends` at the action position; `head_latents` at positions 1–4, predicting the four codes autoregressively (Supplementary Fig. S2). During imagination, codes are sampled from `head_latents`, decoded to a frame and fed back; the actor–critic acts on decoded frames. Figure 1 overviews the model and our instrumentation.

3.2 Agent training and instrumented rollouts

We trained one agent with the public reference implementation and unmodified Crafter hyperparameters (1,000 epochs; 10^7 steps; seed 0) on one NVIDIA H200 (≈ 8.7 days). Over 500 fresh episodes the mean return is 16.20 (s.d. 1.62) with 17.1 achievements unlocked, within one seed-level s.d. of the published 17.8 ± 1.4 [6]. Coverage is dense (18 of 22 achievements unlock in $\geq 73\%$ of episodes); `collect_diamond`, `eat_plant` and the two iron tools unlock too rarely and are excluded from per-achievement statistics. The training pipeline discards environment info, so we re-rolled the frozen agent recording, per step, the four codes, action, reward and 22-bit achievement state (whence just-unlocked indicators): 500 episodes (294,916 steps) for association statistics, and 150 episodes (81,919 steps) with stored frames for activation extraction.

3.3 Association statistics

For every (slot, code, achievement) cell with co-occurrence support ≥ 5 we compute $P(u^i \mid c^k{=}j)$, lift over base rate, and an exact binomial p-value; 190 of 431 cells survive Bonferroni correction. Codebook utilisation is computed by re-encoding the full buffer.

3.4 Isolating the delta trick

We train two baselines on the same buffer and run the identical code→achievement analysis over a common 150-episode set: a *frame-only ablation* (the Δ-IRIS convolutional backbone, 1024 codes, four tokens, but the encoder sees only x_{t+1} and the decoder reconstructs from codes alone) and the *original IRIS* tokenizer (512 codes, 16 tokens/frame, no action). On a common 8,192-transition held-out batch, matched-protocol reconstruction gives PSNR 36.7 (Δ-IRIS), 18.4 (frame-only) and 22.5 (IRIS) with codebook perplexity 80.7, 48.5 and 109.2; Δ-IRIS's lower error is expected, as it reconstructs the residual given (x_t, a_t) (Supplementary Table S3).

3.5 Probes and concept activation vectors

For each timestep we extract seven representations: raw_codes (mean of four codebook vectors); frame_emb (CNN over x_t); wm_latents_emb (the model's code embedding); and the four transformer stages (blocks 0–2 and the final post-norm state), each summarised by averaging the four latent-token positions. Logistic-regression probes for 47 targets (action; instantaneous, and within-5-step reward; just-unlocked and cumulative achievement indicators) use episode-disjoint 80/20 splits. CAVs are the standardised weight vectors mapped back to raw activation space. For the HUD margin we bootstrap the held-out AUROC (B=1000) and report paired CIs on the frame_emb−block difference per concept (Supplementary Table S4).

3.6 Sparse autoencoder

A TopK SAE [12] (M=2048 features, k=16, tied init, unit-norm decoder) is trained on the 81,919 block-1 summary activations (80 epochs, Adam, lr 5×10^{-4}). Each feature is labelled by best-matching achievement (lift within its top activation quartile), CAV (decoder–CAV cosine) and code; five features satisfy strict triple-corroboration. A robustness sweep over widths $M \in \{1024, 2048, 4096\}$, sparsities $k \in \{8, 16, 32\}$, two seeds and four layers gives FVU 0.036–0.054 with $\leq$ 28 dead features; detector features reappear as each variant's own top detector (mean matched cosine 0.63) (Supplementary Table S5). Residual-stream SAEs transfer to vision transformers [28]; their privileged interpretability over simpler bases is debated [29].

3.7 Projection ablation, steering and the empirical null

At the block-1 output we project out a unit direction $\hat{d}$ (matched SAE decoder column, matched CAV, or random) at the four latent positions of a chosen step, and measure $\Delta \log P$ (baseline−ablated) of the four realised codes at the target step, over n=80 moments per cell (gap 0 and gap 3) with 95% bootstrap CIs. For the empirical null we draw R=50 random directions and report the matched effect's percentile/z. Inside imagination we burn in 8 real steps, force the real action, then sample 12 steps with $\hat{d}$ continuously projected out, scoring detector-code occurrence (n=20). Steering replaces projection with addition ($h \mathrel{+}= \alpha\hat{d}$, α up to 8 RMS) from ordinary moments (Supplementary Table S7).

3.8 Multi-direction subspace ablation

We replace single-direction projection with removal of a rank-r orthonormal basis $Q \in \mathbb{R}^{D\times r}$, $h' = h - QQ^\top h$, sweeping $r \in \{1, 2, 4, 8, 12, 16\}$ for eight dominant concepts. The matched CAV subspace is a Gram–Schmidt basis over the concept's CAV and its nearest tech-tree-neighbour CAVs; the SAE-feature stack is its top-r decoder columns by lift; the random subspace is a QR of Gaussian noise. We measure single-step $\Delta \log P$ (gap 0) and, separately, the AUROC of a probe re-fit on activations after projecting out the rank-r CAV subspace.

3.9 Direct logit attribution

For each concept we attribute the `head_latents` logit of its top detector code to a candidate direction by gradient×activation, $\sum_{\text{latent pos}}(\partial\,\text{logit}/\partial\langle h, \hat{d}\rangle)\langle h, \hat{d}\rangle$, with a random other code as control; we report mean absolute detector attribution and detector-vs-control specificity over 18 concepts (Supplementary Table S8).

3.10 Causal tracing

For each unlock window (clean) we build a corrupted run from an ordinary window of the same episode, restore clean activations one (block, timestep, token-type) cell at a time (token types: frame, action, latents), and measure restoration $(\ell_{\text{patched}} - \ell_{\text{corrupt}})/(\ell_{\text{clean}} - \ell_{\text{corrupt}})$ over n=24 window pairs per concept.

Supplementary information. Supplementary Information (Figs. S1–S4, Tables S1–S6) accompanies this article. Interactive HTML artefacts for every experiment are produced by the released code.

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Declarations

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- Conflict of interest: The author declares no competing interests.
- Ethics approval: Not applicable.
- Data availability: Result summaries are included with the code repository; the trained checkpoint and replay buffer (∼170 GB) are available from the author on reasonable request.
- Code availability: All analysis code, Slurm pipelines and figure scripts are at https://github.com/rxailab/delta-iris-interpretability.
- Author contribution: R.X. designed and performed the experiments and wrote the manuscript.

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